

Emotions in Student–LLM Conversations During a Hackathon

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Abstract—Hackathons are competitive events that allow participants to be stimulated by a common goal. There is a time limit and a final ranking, similar to a school exam. In this kind of stressful context, emotions are crucial as they can have a significant impact on participants’ learning capabilities and performances, compared to a more traditional setup. In this article, we analyse the dynamics of emotions during an 11-hour machine learning hackathon, based on 73 student–LLM conversations collected during the event. We do this by using another dataset, StudyChat, as a point of comparison. Emotions were annotated turn by turn by three different LLMs combined through majority voting; we use the word "emotion" instead of "state" for readability. We find that engagement, confusion, and frustration are the three most frequent emotions. Engagement is typically associated with code generation, while confusion and frustration are associated with debugging. We also find that negative emotions are more frequent in the hackathon, while in StudyChat most emotions are more often associated with learning-centered dialogue acts.

Index Terms—hackathon, emotions, education

I. INTRODUCTION

LLMs play a huge role in today’s society. It has been reported that 34% of U.S. adults have used ChatGPT in 2025, about double the share in 2023 [1]. They can be used to learn, complete tasks, or even casually chat. In this paper, we are going to focus on the impact of those tools in education. Several studies have shown that LLM’s impact on the learner can appear in various ways: sustaining engagement, supporting motivation, and even providing emotional support during learning [2]. They are also used across many contexts, from the home to the examination room, thanks to their speed and ease of use. However, they can also harm the user. Every tool requires skills to be used effectively, and with LLMs, using them without understanding can hurt learning rather than help it. Studying how students interact with LLMs is one way to see what goes wrong. In this paper, we look at one signal in particular: emotions.

Emotions and their impact on the user can vary widely from one situation to another. For example, time pressure [3] or interacting with others [4] can intensify them. It is therefore important to know the context in which students are working to better interpret the results. Hackathons are one such context: time-bounded, team-based events where participants try to learn and build in a limited amount of time [5], which makes

them stressful settings. Studying emotions in such a setting can tell us how affect unfolds when students work under pressure.

We therefore provide, in this article, an analysis of emotions in an 11-hour hackathon, organised at an engineering school. Our goal is to better understand the participants’ emotions during the event, a topic that remains under-analysed in the literature. We will first analyse the hackathon’s data before putting it in perspective with StudyChat, a different dataset whose data were collected in more traditional classroom settings, which helps isolate what is specific to the hackathon.

II. RELATED WORKS

A. Hackathons

Most of the literature focuses on the event itself and its purpose [6]–[8], often overlooking the participants. When participants are studied, the focus tends to be on collaboration [9]. The individual behaviour of students therefore remains under-explored.

B. Emotions in Education

Several researchers have studied emotions in a general context. Russell [10], for instance, describes affect along two axes, valence (how pleasant or unpleasant the emotion is) and arousal (how activating or deactivating it is), as shown in Fig. 1. Crossing these two dimensions produces four groups: positive activating (e.g., curiosity, hope, pride), positive deactivating (e.g., relaxation, relief, contentment), negative activating (e.g., frustration, anxiety, anger), and negative deactivating (e.g., boredom, hopelessness, sadness). Some emotions, like confusion, can fall into different groups depending on the situation.

Here we focus on education. A central figure of this field is Pekrun, whose control-value theory [11] explains achievement emotions through two appraisals: control, how much the student feels able to influence the activity and its outcome, and value, how important that activity or outcome is to them. He also provides a precise framework of epistemic emotions (related to knowledge) which includes curiosity, confusion, frustration, and delight.

However, Pekrun’s work relies on questionnaires, not on conversations. In the 2010s, D’Mello and Graesser proposed a framework for conversational settings, derived from students

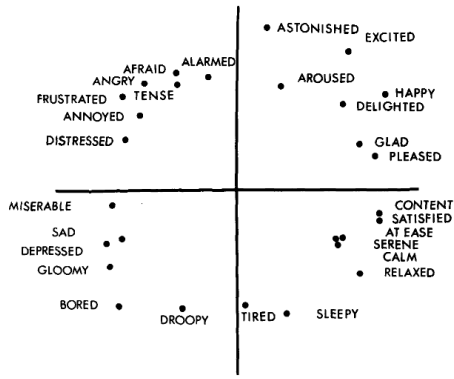


Figure 1. Russell’s circumplex model [10]. Valence on the x-axis, arousal on the y-axis.

interacting with tutoring systems. They first modelled the persistence of emotions [12], and then the transitions between them [13]. Four main emotions are involved: **engagement**, **confusion**, **frustration** and **boredom**. The transitions are illustrated in Fig. 2.

One thing to note is that they identified emotions from multimodal signals, such as facial expressions, not from text. Some emotions, like boredom, are therefore hard to catch in a textual conversation: a bored student simply stops writing.

The link between emotions and performance is also of interest. With the exception of research on anxiety/achievement relations [14], [15], empirical findings on the relationship between other emotions and academic achievement remain limited [16]. Pekrun [11] proposed a model in which emotion/achievement relations are mediated by several variables: cognitive resources, motivation, strategy use, and self-regulated learning. Specific emotions are assumed to impact these variables, which in turn predict performance. According to this model, the emotion/performance relationship is **linear and low-to-moderate** in magnitude. Furthermore, he used Russell’s model, shown in Fig. 1, to characterize emotions. Positive activating emotions have a positive effect on performance, while negative deactivating ones have a negative effect; the two remaining groups have more ambivalent effects.

C. Student–LLM Interactions

This domain is still new, since LLMs appeared recently to the public. One of the main subjects is looking at what students do in the conversations: the actions. McNichols et al. [17] proposed a dialogue act taxonomy derived from student-LLM chat logs collected during a university-level AI course. Their taxonomy is summarised in Table I.

III. METHODOLOGY

A. Data

In this article, we use two main data sources: the hackathon and StudyChat [17], a publicly available dataset of real-world student interactions with an LLM-powered tutoring chatbot (ChatGPT 4.1). Thanks to its difference in context, StudyChat

Table I
DIALOGUE ACT TAXONOMY, ADAPTED FROM McNICHOLS ET AL. [17].

Dialogue act	Definition/Example
WRITING	
WRITE CODE	Generate new code
WRITE ENGLISH	Write prose (report, comment)
CONVERSION	Convert code or data to another form
SUMMARIZE	Summarize a text or an output
OTHER	Other writing request
EDITING	
EDIT CODE	Modify existing code
EDIT ENGLISH	Modify existing prose
OTHER	Other editing request
CONCEPTUAL QUESTIONS	
PYTHON LIBRARY	Question about a Python library
PROG. LANGUAGE	Question about the language itself
COMPUTER SCIENCE	Question about a CS or ML concept
PROG. TOOLS	Question about a tool (git, jupyter)
MATHEMATICS	Question about a maths concept
OTHER	Other conceptual question
CONTEXTUAL QUESTIONS	
ASSIGN. CLARIFICATION	What the task expects
CODE EXPLANATION	What a specific snippet does
INTERPRET OUTPUT	How to read a specific result
OTHER	Other contextual question
PROVIDE CONTEXT	
ASSIGN. INFORMATION	Pastes the task description
ERROR MESSAGE	Pastes a console error
CODE	Pastes code without a question
OTHER	Other context provided
VERIFICATION	
VERIFY CODE	Is this code correct?
VERIFY REPORT	Is this text correct?
VERIFY OUTPUT	Is this result correct?
OTHER	Other verification request
OFF TOPIC	
CHIT-CHAT	Casual talk
GREETING	Hello, goodbye
GRATITUDE	Thanks
OTHER	Other off-topic message
MISC	
OTHER	-

acts as a point of comparison to see what is specific to the hackathon. Table II gives an overview of both.

The conversations were made public by the participants and collected for analysis. The uploaded images were excluded.

B. Framework of Emotions

Identifying the framework of emotions that we are going to use is important, as it provides a clear list for classification. However, not all emotions are suited to our context. We are dealing with a conversation, in which people are trying to learn and to perform a task.

We chose D’Mello and Graesser’s model because their emotions come from students interacting with conversational tutoring systems, which is the same kind of data we work with,

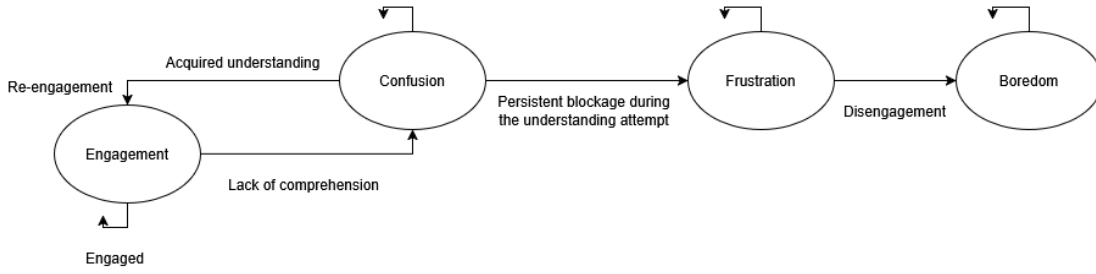


Figure 2. Emotions and their transitions. Adapted from D’Mello and Graesser [12], [13].

Table II
OVERVIEW OF THE TWO DATASETS.

Property	Hackathon	StudyChat
Number of students	73	203
Total turns (messages)	1825	16851
Conversations	73	2214
Avg. turns per conversation	24.9	7.6
Language	French	English
Duration	11 hours	2 semesters
Task	Coding competition (Titanic / Kaggle)	Python assignments (AI course)
Time pressure	Yes	No

while Pekrun’s scales rely on questionnaires. We also added two other labels: neutral, which indicates a lack of affect, and curiosity, which belongs to D’Mello and Graesser’s broader set of learning-centered emotions but not to their transition model.

D’Mello and Graesser refer to these labels as affective states, since engagement and boredom are not emotions in the strict sense. For readability, we use the term *emotions* throughout this paper.

Table III gives the definition of each emotion, along with generic examples and a clarification of the boundaries between them.

C. Tools Used for Annotation

The tools used for annotation are the LLMs themselves. Indeed, both datasets mix natural language with code and error traces, and the labels are learning-centred emotions that depend on the conversational context rather than on isolated messages. HuggingFace classifiers are generally trained on generic emotions (joy, pride, etc.) and single messages. The LLM solution was therefore the most logical one. They enable consistent labelling at a scale infeasible for human annotators, and have been shown to match or exceed crowd-workers on several text-classification tasks [18], [19].

However, all LLMs carry biases coming from their training data, so it is necessary to adopt a multi-LLM approach in order to reduce the impact of these biases: for each message, three LLMs vote, and the chosen result is the most voted one. We did two separate annotations, one for the emotions and one for the actions.

The models were mainly selected according to an emotion benchmark, as annotating actions is a simpler task: EQ-Bench [20] (Emotional Intelligence Benchmark for LLMs). The chosen models are close to one another in performance: **Mistral-Nemo**, **Qwen 2.5-14B**, and **Gemma-9B**. If the three models give three different results, the chosen answer is that of the highest-ranked model in the benchmark.

The models annotated emotions in both datasets, but actions only in the hackathon, as StudyChat already provides dialogue-act labels (generated by GPT-4.1). We aligned our action taxonomy with theirs to allow comparison. We wrote our own prompts for both emotion and action annotation, using the same emotion prompt across the two datasets. They are available in our repository (in French).¹ However, since StudyChat’s dialogue acts were produced with a different model and prompt, action distributions across the two datasets should be compared with caution.

We used two different context windows depending on the task. For emotions, we set N=16. This means that the sixteen previous student prompts and the sixteen corresponding LLM responses are included (32 messages in total). We first tried using the full conversation as context, but this produced unstable outputs on the longest conversations (invalid labels and malformed responses). To set N, we ran an ablation over window sizes (Table V). Majority agreement between the three models stays flat across all windows (around 88%), so a larger window costs nothing. Labels also change less and less as the window grows. We therefore keep N=16. Beyond it, larger windows change the labels only marginally while adding context cost. For actions, however, we took the full conversation, in order to align with the methodology of McNichols et al. [17], who did not justify this choice in their article. We think their decision is based on the average number of turns per conversation in their dataset, which is considerably larger than in ours (24.9 vs. 7.6). This gives them more room to leverage context.

D. Transitions

Our first idea was to compute transitions using only conditional probabilities, without taking the base rates into account:

$$\Pr(M_{t+1} | M_t) = \frac{\Pr(M_{t+1} \cap M_t)}{\Pr(M_t)} \quad (1)$$

¹<https://github.com/batg099/hackathon>

Table III
EMOTIONS, ADAPTED FROM D’MELLO AND GRAESSER [12].

Emotion	Definition	Critical boundary	Example
ENGAGEMENT	The task is progressing	vs. NEUTRAL: continuity with prior turns	“ok that works, now add a filter on the date”
CONFUSION	An impasse is met during the task	-	“why does this return undefined?”
FRUSTRATION	The impasse is not resolved, or a goal is blocked	vs. CONFUSION: repetition	“still the same error, I already tried that”
BOREDOM	Frustration left unresolved: disengagement	-	“whatever, just give me the whole file”
CURIOSITY	Exploration beyond the immediate need	vs. CONFUSION: no impasse, the task is not blocked	“out of curiosity, why is pandas faster here?”
NEUTRAL	No emotion detected	-	“import matplotlib”

Table IV
AGREEMENT BETWEEN THE MODEL VOTE AND HUMAN ANNOTATION ON 180 TURNS (COHEN’S $\kappa = 0.194$). CURIOSITY, BOREDOM AND NEUTRAL WERE NEVER ASSIGNED BY THE HUMAN ANNOTATOR ON THIS SAMPLE AND ARE OMITTED.

Emotion	P	R	F1	n
CONFUSION	0.30	0.55	0.39	29
FRUSTRATION	0.18	0.58	0.28	12
ENGAGEMENT	0.92	0.47	0.62	139
Macro avg	0.28	0.32	0.26	180

Table V
CONTEXT-WINDOW ABLATION ON HACKATHON CONVERSATIONS (≥ 12 TURNS). *Instability*: MEAN % OF LABELS THAT CHANGE FROM THE PREVIOUS WINDOW, AVERAGED OVER THE THREE MODELS (LOWER = MORE STABLE). *Agreement*: % OF TURNS WITH UNANIMOUS (3/3) AND MAJORITY ($\geq 2/3$) LABELS AT THAT WINDOW.

Window	Instability	Agree. (3/3)	Agree. ($\geq 2/3$)
2	-	45.2%	87.9%
4	24.6%	45.3%	88.5%
6	16.3%	44.2%	88.2%
8	12.4%	44.4%	87.8%
12	14.2%	41.3%	87.2%
16	10.0%	39.6%	86.6%
20	8.4%	38.0%	86.8%

However, conditional probabilities are biased by the base rate of M_{t+1} : if an emotion is very frequent, the probability of transitioning to it will be high regardless of any genuine association between M_t and M_{t+1} . To address this, Baker et al. [21] used the following likelihood metric:

$$L(M_t \rightarrow M_{t+1}) = \frac{\Pr(M_{t+1} | M_t) - \Pr(M_{t+1})}{1 - \Pr(M_{t+1})} \quad (2)$$

L is positive when the association is stronger than chance, zero when the two states are independent, and negative otherwise.

Following D’Mello and Graesser’s methodology [13], we studied transitions between *distinct* states. Each sequence was recoded to remove consecutive repetitions ($X \rightarrow Y \rightarrow Y \rightarrow Z$ becomes $X \rightarrow Y \rightarrow Z$), so **self-transitions are excluded**. Transitions were counted within conversations only.

We computed L per participant, then ran a one-sample t -test against zero for each transition. A transition was considered reliable when its mean L was significantly positive ($p < .05$).

E. Student Performance

Each participant received a grade at the end of the hackathon. Since participants worked in groups, every member of a group was assigned the same grade, and each participant was then classified into one of three performance tiers: **low** (bottom 25%), **medium** (middle 50%), and **high** (top 25%).

IV. RESULTS

In Fig. 3, engagement decreases with performance: low and medium students are engaged around half the time, high students less (median 0.35) and in a tighter range. Confusion goes the other way, highest for high-level students. Frustration is near zero for medium students but clearly present for high ones, who show the widest frustration range of the three groups.

In Fig. 4, we find that engagement, confusion and frustration are the most frequent emotions for both datasets (neutral not being an emotion). Positive emotions are higher in StudyChat, while negative ones are higher in the hackathon. Frustration is the clearest case: in the hackathon it is present in most conversations (median 0.09), whereas in StudyChat it is absent from the majority but concentrated in a few (median 0, with outliers up to 0.65). The hackathon distributions are also tighter, with fewer values at the extremes.

In Fig. 5, the two datasets show similar transition patterns overall. In both, engagement $\rightarrow$ confusion ($L = 0.28$ and 0.28), confusion $\rightarrow$ engagement (0.26 and 0.25), and confusion $\rightarrow$ frustration (0.14 and 0.08) are reliable ($p < .05$). Frustration $\rightarrow$ engagement is present in both datasets (0.11 and 0.14). Neutral $\rightarrow$ engagement (0.30 and 0.44) and curiosity $\rightarrow$ engagement (0.51 and 0.41) are the strongest off-diagonal transitions.

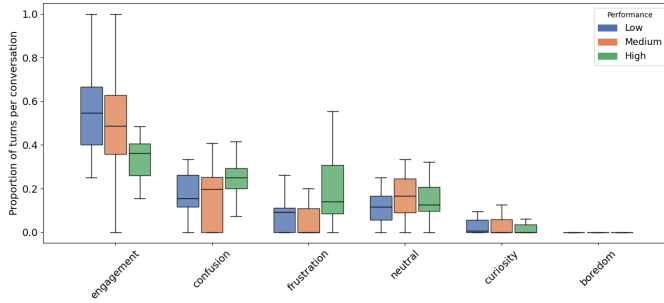


Figure 3. Distribution of emotions in the hackathon, per performance.

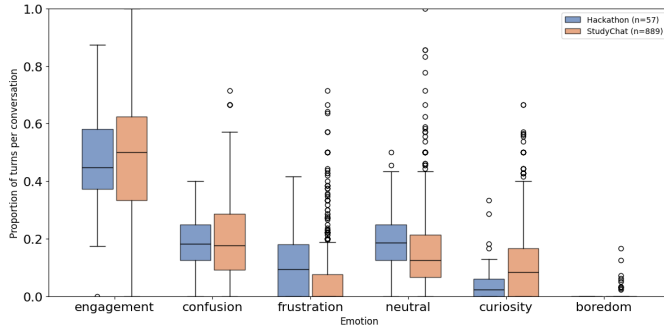


Figure 4. Distribution of emotions in the hackathon and StudyChat. The x-axis represents the emotions and the y-axis the proportion of turns per conversation (between 0 and 1). Only conversations with more than five turns are included.

In Fig. 6, most students across both datasets mostly generate and debug code. However, there is a clear distinction in most emotions. In StudyChat, they are more associated with learning-centered actions (asking conceptual questions) than in the hackathon.

V. DISCUSSION

We obtain high majority (2/3) agreement between models (86%), indicating that labelling is mostly consistent across them. However, this consistency does not imply validity. To check this, we compare the LLM labels against human annotations using Cohen’s kappa (κ), a measure of inter-rater agreement, which quantifies agreement on categorical labels while correcting for chance ($\kappa = 1$ being perfect agreement, 0 chance level). We obtain a low kappa score, meaning that agreement between human annotators and LLMs is low. We asked the annotator where they struggled, and their answer was clear: transitions were still not clear. One example is confusion to frustration: if we consider "time" as a decisive factor, how do we know when a student is frustrated? How exactly can we measure "time" in a conversation? Annotating conversations of students doing work is different from doing it in regular conversation contexts like friendly messages. Low agreement is expected for these situations. Bosch et al. [22] report $k = 0.22$ for confusion and 0.23 for frustration using the same framework. Their numbers come from facial data, not text, so the comparison is only indicative, but it suggests these emotions are intrinsically hard to annotate.

Linking emotions and performance is important, as it gives an idea of how one influences the other. Our first idea was to correlate transitions with performance. However, to our knowledge, few studies correlate affective transitions with academic performance. Most of them look at the final emotion levels. We choose to do the same. We also did not rely on Pekrun’s predictions based on Russell’s model, which would have been a valuable but complex addition. For all three types of students, the same pattern holds (Fig. 3): students are engaged most of the time, sometimes confused, and occasionally frustrated. This distribution is consistent with D’Mello’s meta-analysis of 24 studies [23]. However, emotions like boredom and curiosity are nearly non-existent. Boredom doesn’t suit conversational annotation, and curiosity is a contextless emotion, which is a problem, as they are performing a task that relies on its context. Frustration is the key emotion. Based on Ruiz-Ortega and Berrios-Martos [24], frustration intolerance is linked to performance. Our hypothesis is that the level of frustration doesn’t matter; what does is how students handle it. Following D’Mello and Graesser’s model [13], confusion and frustration can help learning when they are resolved, not when they last. We assume that what helps learning also helps performance, we therefore expect top students not to show less frustration, but a stronger tendency to exit it, and low-performing students to remain in frustration longer.

Next, we analyse the distribution of emotions for both datasets in order to have a first idea of what is happening. We expect higher levels of negative emotions in the hackathon, as time pressure, the competitive task, and the final ranking together make it a more stressful setting than StudyChat. One difference is unrelated to stress, though: language (French vs English). These results should therefore be read with caution. Translating would have removed this gap but introduced a translation bias of its own. Our results in Fig. 4 could be explained by the fact that stressful events trigger negative emotions [25]. Also, students in hackathons are engaged in an active task and under time pressure, which could explain the fact that students ask more conceptual questions in StudyChat.

We then analyse the transition matrices for both datasets (Fig. 5), using D’Mello and Graesser’s work as a reference [13]. The core of their model is reproduced in both datasets. The engagement $\rightarrow$ confusion transition is reliable (0.28 and 0.28), consistent with the disequilibrium hypothesis, as is confusion $\rightarrow$ engagement (0.26 and 0.25), reflecting productive confusion, and confusion $\rightarrow$ frustration (0.14 and 0.08), reflecting persistent confusion. The model suggests that frustration $\rightarrow$ engagement should be absent. In the Hackathon it is indeed not reliable (0.11), but in StudyChat it is reliable (0.14), suggesting that learners there return to engagement after frustration rather than remaining stuck. The time pressure of the Hackathon may make this recovery harder. Neutral and curiosity were not taken into consideration in their work. According to Fig. 6, neutral is linked to debugging in the Hackathon and to conceptual questions in StudyChat. In StudyChat, neutral is followed by engagement (0.44), consistent with neutral there consisting of conceptual questions

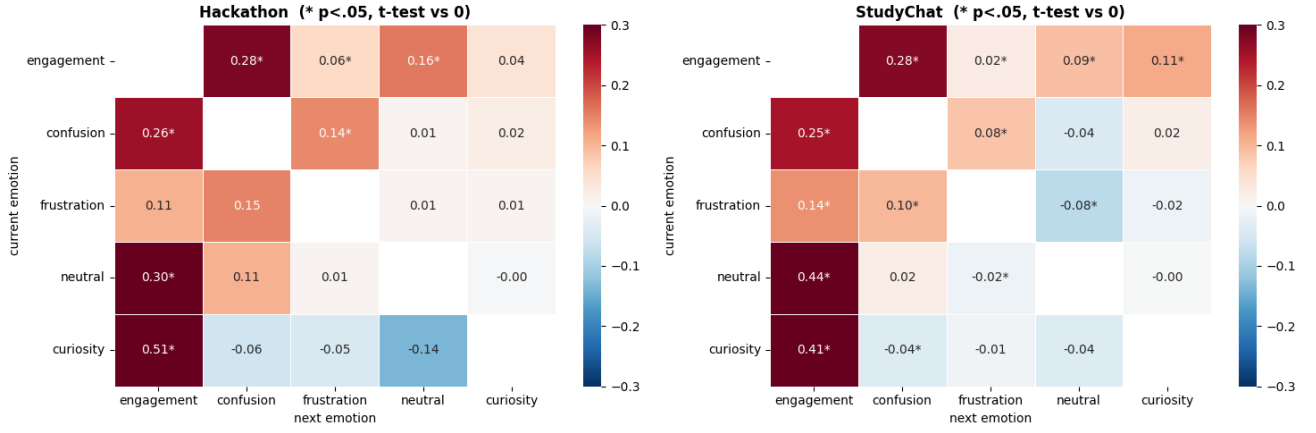


Figure 5. Transition matrices for the Hackathon and StudyChat datasets (Eq. 2). The diagonal is null by construction (consecutive repetitions removed). Cells show the mean likelihood L across subjects; asterisks mark transitions significantly above chance (one-sample t -test, $p < .05$).

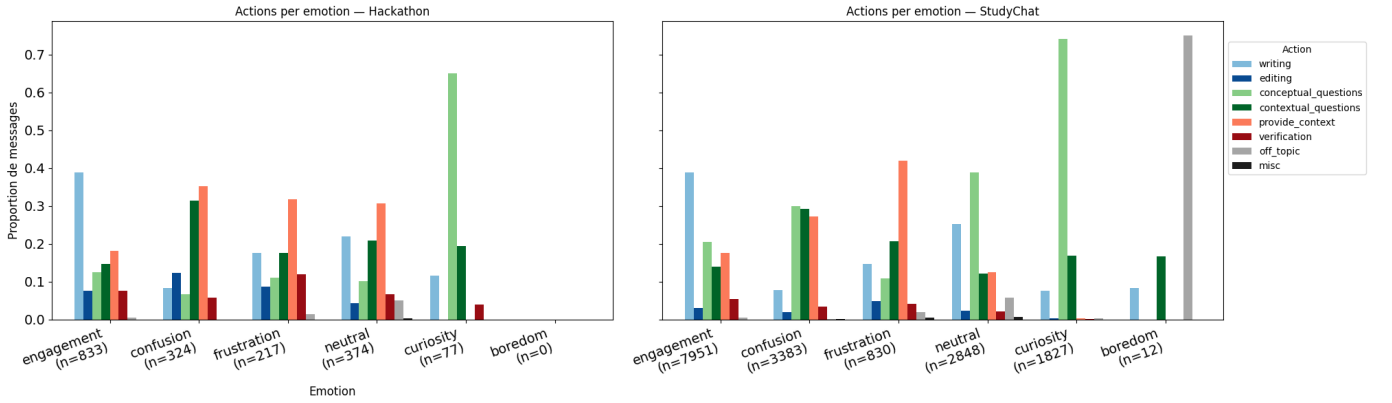


Figure 6. Actions per emotion across the hackathon and StudyChat datasets.

that keep students engaged. We also think that the neutral $\rightarrow$ confusion tendency in the Hackathon could be explained by debugging, as participants there face more code problems.

VI. ETHICAL CONSIDERATIONS

This study was conducted in accordance with the EU General Data Protection Regulation (GDPR) and the French Data Protection Act of 6 January 1978, the data collection relying on the research exemption set out in Article 89 of the GDPR. Every student included in the dataset gave their free and informed consent to the collection and processing of their conversations for research purposes only, after being clearly informed of the study’s procedures. Participation was voluntary, and participants retained the right to access, rectify, object to, restrict, or request the deletion of their data at any time, as well as to choose whether their data would be destroyed or archived for future scientific use. A named Data Controller and a Data Protection Officer were made available to handle any such request. All personally identifiable information was removed prior to analysis.

VII. FUTURE WORK

A key limitation is that we cannot measure the evolution of learning/performance. Without a measure of students’ initial level, any association between interaction profile and grade remains correlational: we cannot determine whether a behaviour improves performance, or whether stronger students simply behave that way. Future data collection could address this in two ways. Organising several hackathons with the same students would provide repeated performance measures, allowing prior scores to be used as a control. Adding a short theoretical exam before and after each event would measure learning rather than performance alone, separating prior knowledge from what students actually learned.

VIII. CONCLUSION

In this paper, we analysed emotions in 73 student-LLM conversations collected during an 11-hour hackathon, using a three-LLM majority vote and StudyChat as a comparison point. We found that engagement, confusion and frustration dominate both settings. Engagement is typically associated with code generation, while confusion and frustration are associated with debugging. We also find that negative emotions

are more frequent in the hackathon, while in StudyChat most emotions are more often associated with learning-centered dialogue acts. These findings remain preliminary, given the low human-model agreement ($\kappa = 0.194$).

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